

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 05

Configure Tiered AWS Billing Alerts and Email Notifications

AWS Billing | CloudWatch | Amazon SNS | Email Alerts

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
RELEASE DATE	July 8, 2026
STATUS	Complete

Executive Summary

Enabled AWS Free Tier alert delivery and CloudWatch billing alerts, then created tiered cost thresholds at **\$10**, **\$25**, and **\$50**. All CloudWatch billing alarms were linked to the existing **SecurityAlerts** Amazon SNS topic, and the topic was configured with a confirmed email subscription so future billing alerts are delivered to a monitored inbox.

Business Problem

Unexpected cloud spending can indicate either an operational mistake or a security incident. Forgotten resources, compromised credentials, cryptomining, unusual data transfer, and other forms of abuse can all increase account charges. The environment needed multiple warning thresholds and a reliable delivery path so cost anomalies would be noticed before they became large.

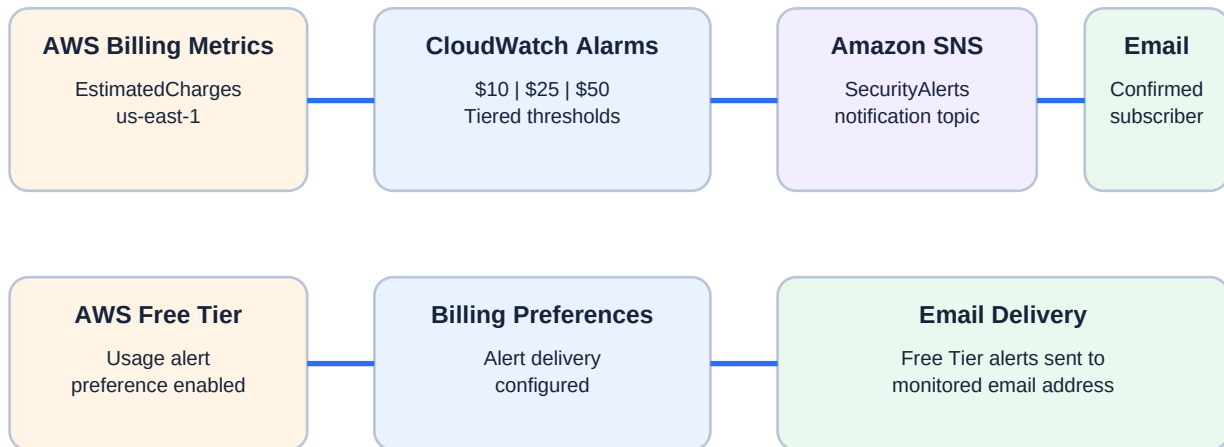
Objectives

- Enable AWS Free Tier usage alert delivery.
- Enable CloudWatch billing alerts.
- Create tiered billing alarms at \$10, \$25, and \$50.
- Configure alarms in us-east-1, where AWS billing metrics are available in CloudWatch.
- Link every billing alarm to the SecurityAlerts SNS topic created in Project 04.
- Create and confirm an email subscription to receive SecurityAlerts notifications.

Environment

Cloud Provider	Amazon Web Services (AWS)
Primary Monitoring Service	Amazon CloudWatch
Billing Metric Region	us-east-1 (N. Virginia)
Metric	AWS/Billing - EstimatedCharges
Thresholds	\$10, \$25, and \$50
Notification Topic	Amazon SNS - SecurityAlerts
Subscription Protocol	Email
Free Tier Alerts	Enabled
CloudWatch Billing Alerts	Enabled

Architecture



Billing thresholds feed the existing SecurityAlerts SNS topic, while Free Tier usage alerts are delivered directly through billing preferences.

Figure 1 - Billing and Free Tier alert notification architecture.

Implementation Summary

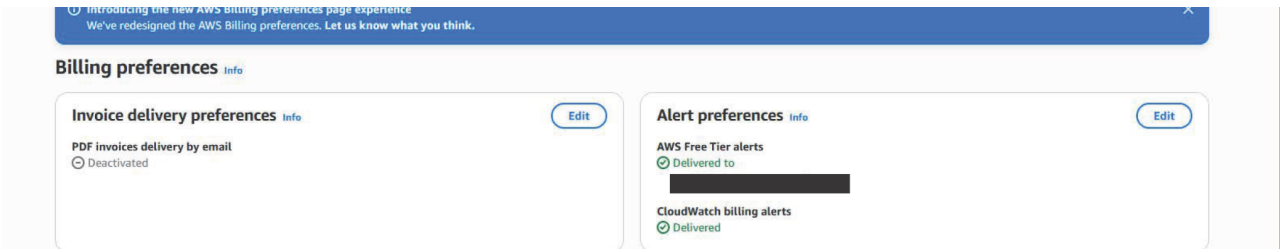
1. Enabled AWS Free Tier alert delivery in Billing preferences.
2. Enabled the setting required to receive CloudWatch billing alerts.
3. Opened CloudWatch in us-east-1 and selected the AWS/Billing EstimatedCharges metric.
4. Created tiered billing alarms for \$10, \$25, and \$50 estimated monthly charges.
5. Linked each CloudWatch alarm action to the existing SecurityAlerts SNS topic.
6. Created an email subscription to SecurityAlerts and confirmed the subscription.
7. Validated the billing alarm list, alert preferences, and confirmed SNS email delivery path.

REGION NOTE

The alarms are created in us-east-1 because CloudWatch exposes AWS billing metrics there. This does not mean charges only occur in that region; it is the region used to view and alarm on account-level billing metrics.

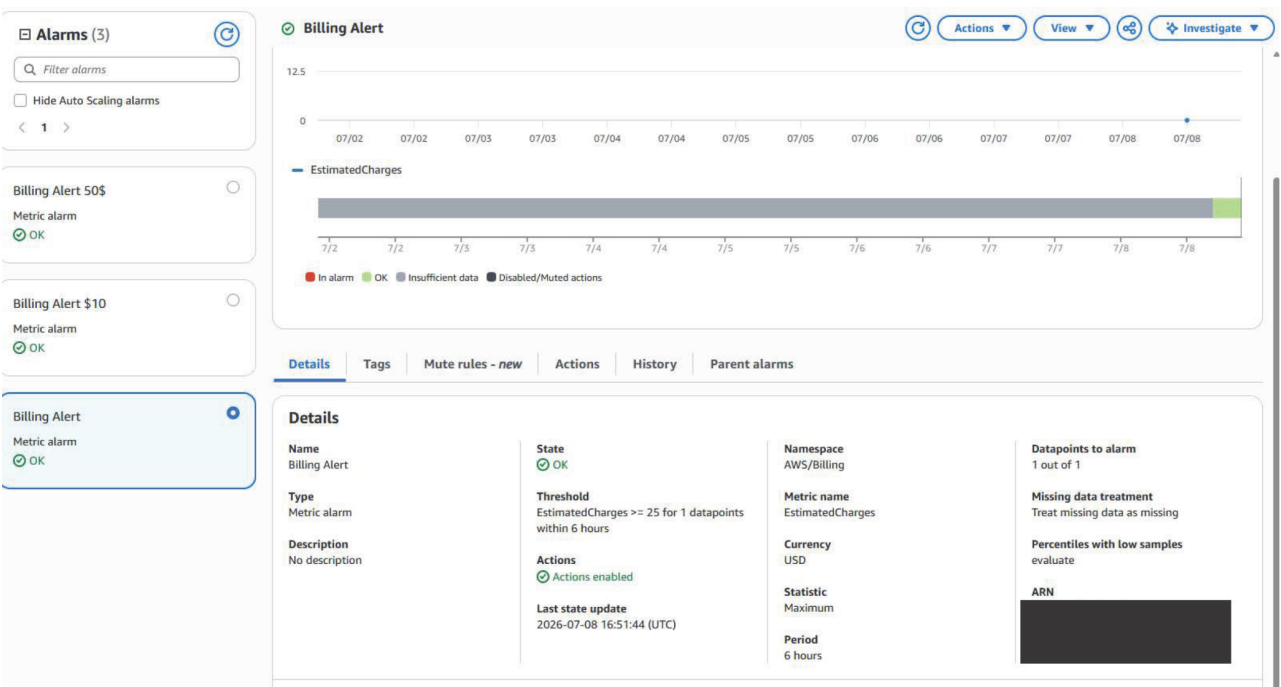
Evidence

Figure 2 - Billing and Alert Preferences



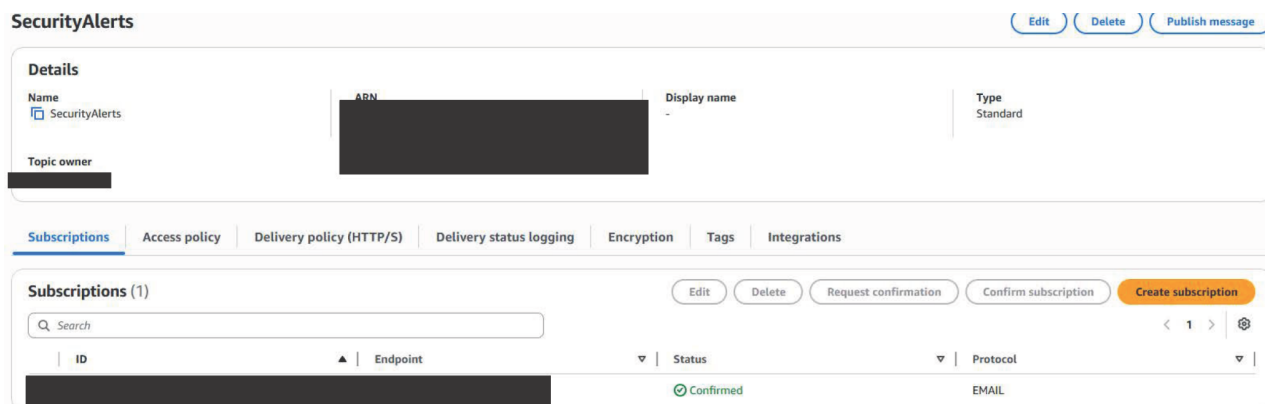
The redacted Billing preferences page confirms AWS Free Tier alerts are configured for delivery and CloudWatch billing alerts are enabled.

Figure 3 - Tiered CloudWatch Billing Alarms



The CloudWatch alarm view shows three billing alarms. The alarm list includes \$10 and \$50 thresholds, while the selected alarm details show an EstimatedCharges threshold of \$25. Together, the evidence confirms the \$10, \$25, and \$50 tiered alert strategy.

Figure 4 - Confirmed SecurityAlerts Email Subscription



The SecurityAlerts SNS topic shows one email subscription in Confirmed status. Sensitive endpoint and topic identifiers were redacted before publication.

Security Analysis

Billing alerts are a detective control. They do not prevent abuse, but they can reveal unexpected consumption patterns that deserve investigation.

Tiered thresholds provide escalating awareness. A low threshold can catch mistakes early, while higher thresholds create additional notification points if costs continue to rise.

Linking the alarms to the existing SecurityAlerts SNS topic centralizes security notifications and creates a reusable delivery path for later monitoring projects.

A confirmed email subscription is essential. An unconfirmed SNS subscription would leave the alerting pipeline incomplete even if the CloudWatch alarms were configured correctly.

Lessons Learned

Cost monitoring is part of cloud security. Unexpected spending can be an indicator of compromised credentials, unauthorized resource creation, or simple operational mistakes.

AWS billing alarms use the EstimatedCharges metric in CloudWatch and must be configured in us-east-1.

Notification architecture matters as much as the alarm itself. The alarm, SNS topic, and confirmed subscriber all have to work together for the control to be useful.

Production Improvements

- Add higher thresholds appropriate to the expected production budget and business risk tolerance.
- Use AWS Budgets and Cost Anomaly Detection alongside CloudWatch billing alarms.
- Create multiple notification subscribers or operational channels to avoid a single point of notification failure.
- Document an incident-response procedure for unexpected billing alerts.
- Test the notification path periodically and review alarm actions after SNS or email changes.

- Consider account- and service-specific cost controls in multi-account environments.

Skills Demonstrated

AWS Billing preferences	Cost anomaly awareness	Amazon SNS
CloudWatch alarms	Detective controls	Email subscriptions
Threshold configuration	Security incident indicators	Alert routing

Resume Bullet

RESUME	Implemented tiered AWS cost monitoring by enabling Free Tier and CloudWatch billing alerts, creating \$10, \$25, and \$50 EstimatedCharges alarms in us-east-1, and routing notifications through a confirmed SecurityAlerts SNS email subscription.
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STAR Interview Story

Situation: My AWS lab environment needed protection against unexpected costs caused by mistakes or suspicious resource usage.

Task: Create an early-warning system that would notify me at multiple spending levels.

Action: I enabled Free Tier and CloudWatch billing alerts, created \$10, \$25, and \$50 EstimatedCharges alarms in us-east-1, linked all alarms to the SecurityAlerts SNS topic, and confirmed an email subscription.

Result: The account now has a tiered cost-monitoring and notification pipeline that can surface unexpected spending before it becomes a larger operational or security issue.

Version History

v1.0.0	July 8, 2026	Initial documented implementation. Enabled Free Tier and CloudWatch billing alerts, created \$10, \$25, and \$50 billing thresholds in us-east-1, linked all alarms to SecurityAlerts, confirmed the email subscription, and embedded all provided project evidence using portfolio documentation standard v1.0.1.

References

- Cloud Security Lab a Week (S.L.A.W.) - Follow the Money!.
- User-provided, redacted CloudWatch billing alarms screenshot.
- User-provided, redacted AWS Billing preferences screenshot.
- User-provided, redacted SecurityAlerts SNS subscription screenshot.

- Project 04 - Deploy Multi-Region Security Alert Topics with AWS CloudFormation.